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SPATIAL DATA ANALYSIS
OPEN ENDED LAB (OEL)

SPATIOTEMPORAL LAND USE/LAND COVER CHANGE DETECTION IN LAHORE (2016–2024) AND FUTURE FORECASTING (2030–2100) USING GOOGLE EARTH ENGINE AND CA-MARKOV MODELS

Abstract: Land Use and Land Cover (LULC) monitoring plays a significant role in understanding how speedily growing cities transform over time. This study analyzes multi-temporal satellite imagery of Lahore for the years 2016, 2018, 2020, 2022, and 2024 using the Google Earth Engine platform. Supervised machine-learning classification was practical to map major land-cover categories including built-up areas, crop land, water, tree cover and bare land. Accuracy assessment was performed using the confusion matrix and Kappa coefficient to ensure consistency of classification outputs [1]. To access future landscape dynamics, the CA–Markov model was employed to simulate long-term LULC scenarios for 2030, 2050, 2070. The results reveal a continuous development of built-up areas at the cost of vegetation and agricultural land, reflecting urban spread and population-driven land depletion. The predicted maps highlight a significant drop in natural cover by the end of the century if current trends continue [4,5]. The study provides essential insights for officials and urban planners to support sustainable land-use strategies in Lahore.

KEYWORDS: Land Use/Land Cover (LULC), CA–Markov Model, Change Detection, Accuracy Assessment, Kappa Coefficient, Random Forest Classifier

1. Introduction

Land Use and Land Cover (LULC) change has become one of the most habitually investigated subjects in remote sensing and geospatial research. In firm expanding urban centers such as Lahore, understanding the arrangement, rate, and drivers of LULC renovation is essential for operational environmental management, infrastructure planning, and source allocation. Population growth, commercial development, and transportation extension are major suppliers to the alteration of natural and agricultural areas into built-up land in Lahore. These changes not only amend the physical landscape but also influence ecological stability, urban heat intensity, groundwater recharge, and overall ecological quality.

Remote sensing offers a profitable and reliable approach for multi-temporal monitoring of landscape underlying forces [2]. The accessibility of free, long-term satellite data such as Landsat and Sentinel-2, tied with the processing capabilities of Google Earth Engine (GEE), has ominously enhanced the accuracy and proficiency of LULC classification. Machine-learning classifiers, particularly Random Forest and CART, have verified strong performance in unrelated urban environments by effectively unique complex land-cover classes [6,7].

In the framework of Lahore, where agricultural land is reliably being replaced by residential, industrial, and commercial developments, LULC prediction becomes essential for sustainable planning [7]. This study therefore aims to: (i) classify LULC for 2016–2024 using machine learning in GEE, (ii) explore temporal change patterns, and (iii) simulate future LULC for 2030–2100 using CA–Markov modelling. The findings provide a methodical

groundwork for policy decisions relating to urban growth, environmental conservation, and land-management schemes in Lahore.

2. Materials

2.1 Study Area

Lahore District, located between $31^{\circ}15'–31^{\circ}45' \text{ N}$ and $74^{\circ}01'–74^{\circ}39' \text{ E}$, is the capital of Punjab and one of Pakistan's most populated regions. Once dominated by agricultural land, Lahore has experienced wide urban expansion due to population growth, commercial activity, and infrastructure development. These factors make Lahore a promising case study for exploring multi-temporal LULC change and future urban growth patterns.

2.2 Data acquisition

Satellite imagery and spatial data applied in this study were retrieved via the cloud-based Google Earth Engine (GEE) platform. Multi-temporal datasets spanning the years 2016, 2018, 2020, 2022, and 2024 were traced from the Landsat 8 Collection 2 Level-2 and Sentinel-2 MSI Level-1C origin within the GEE public data catalog.

3. Work flow diagram

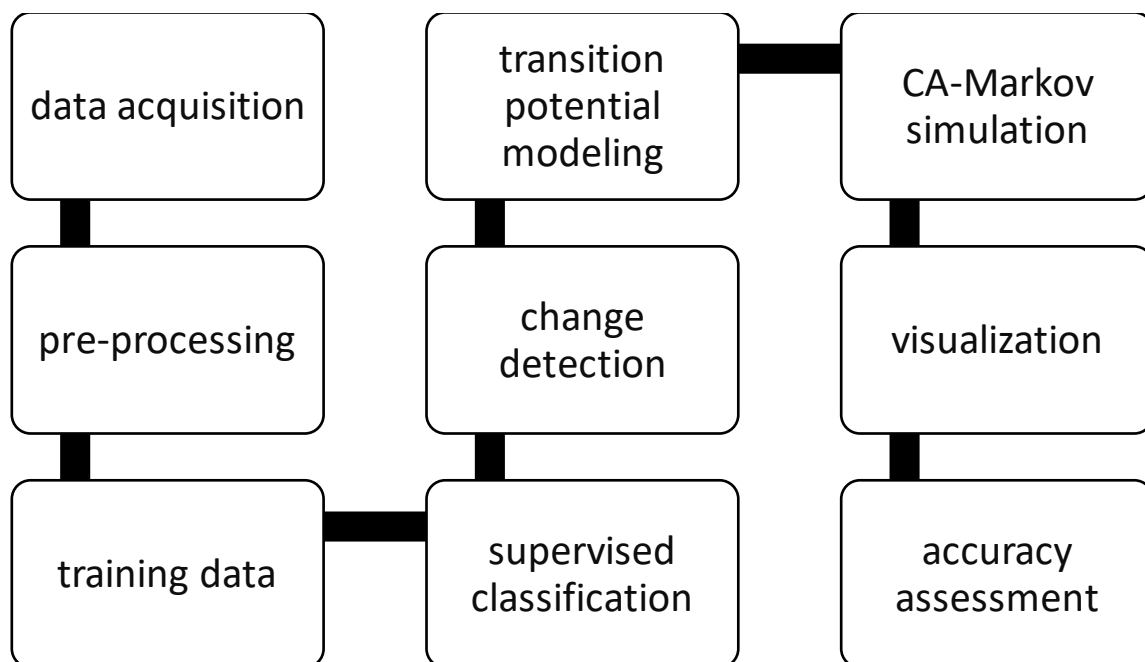


Figure 1. Workflow diagram summarizing the steps of data acquisition, preprocessing, supervised LULC classification, accuracy assessment, change detection, transition potential modelling, and CA–Markov simulation used for future land-cover prediction in Lahore.

4. Methods

4.1 Shape file extraction

Lahore shape file was extract in google earth engine from FAO GAUL simplified level-2 dataset [9,10]. Sentinel-2 Surface Reflectance (S2_SR) was used to attain imagery (1 August to 15 September 2024). Images with less than 10% cloud cover and true visualization.

Add additional shape files for slope, elevation, population density, distance to roads, distance to water and urban-rural.

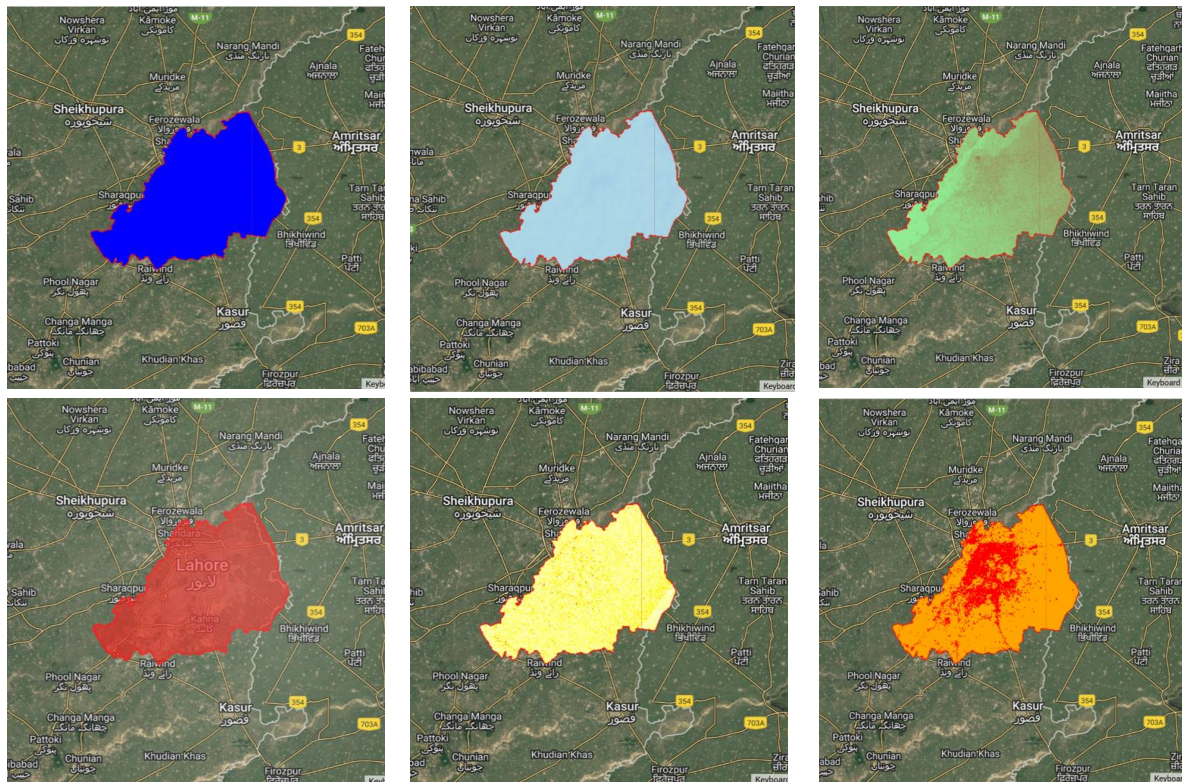


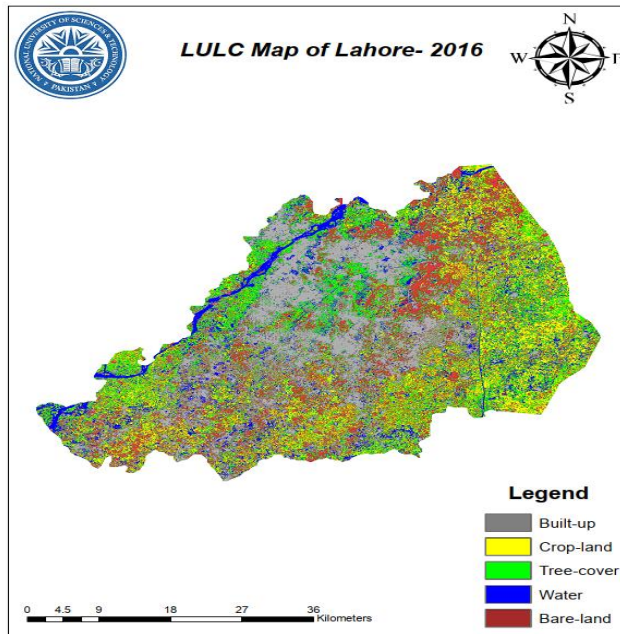
Figure 2. Geographical location and administrative boundary of Lahore District, extracted from the FAO GAUL Level-2 dataset and used as the study area for LULC analysis and modelling.

4.1.1 Selection of classes and training points

The major land cover categories were first identified by visual interpretation. Common classes include built-up land, cropland, water, tree cover, and bare land. Training points were physically implanted inside the Google Earth engine through the drawing tool. Several points were placed across Lahore [5]. The points were spread over a homogenous area.

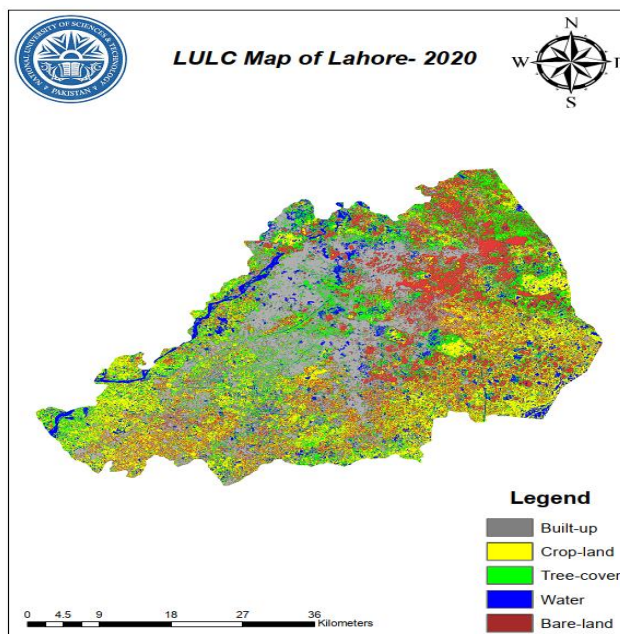
4.2 Land use/Land cover(LULC) classification

All training samples organized for the study were first compound into a integrated dataset to ensure reliable classification across all target years. Earlier to analysis, Sentinel-2 imagery was administered using a cloud-masking algorithm to obtain clear, high-quality prospects suitable for land-cover analysis.



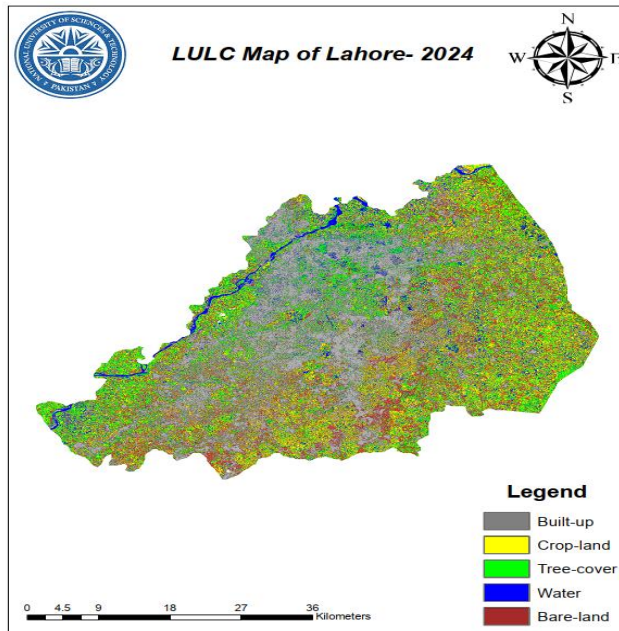
For the year 2016, Landsat 8 surface reflectance imagery from August–September was nominated due to its relatively low cloud presence. Cloud-affected sections were excluded, reflectance values were consistent, and a center composite was produced to represent the seasonal conditions. This composite was ordered using a Random Forest classifier trained on the merged training dataset. The output comprised five major land-cover categories: Built-up, Cropland, Tree Cover, Water, and Bare Land.

Figure 3. Land Use/Land Cover classification map of Lahore for the year 2016 using Landsat 8 imagery, showing spatial distribution of built-up land, cropland, tree cover, water, and bare land.



The same methodological steps were functional to Landsat 8 imagery for 2020. The resulting LULC maps provided an reorganized representation of land-cover surroundings and their dispersal across the district. The classification process was also frequent using the SMILE Random Forest algorithm to validate the sturdiness of the results.

Figure 4. Land Use/Land Cover classification map of Lahore for the year 2020 derived from Landsat 8 imagery, highlighting major LULC transitions compared to 2016.



For 2024, higher-resolution Sentinel-2 imagery (10 m) was hired. Following preprocessing, supervised classification was again done using both Random Forest approaches. The outputs produced thorough land-cover maps and class statistics, offering a refined illustration of recent site patterns in Lahore.

Figure 5. Land Use/Land Cover classification map of Lahore for 2024 generated from Sentinel-2 imagery, providing an updated representation of recent land-cover conditions.

4.3 Land use/Land cover (LULC) change detection

Five land-cover categories ;Built-up, Cropland, Tree Cover, Water, and Bare Land were selected as the classification basis. Each class was allotted a harmonized numeric code (1–5) to maintain consistency in labeling and clarification throughout the analysis.

4.3.1 Area calculation

The spatial level of each land-cover class was instigated by including the number of pixels allotted to each class for the respective years [8]. Pixel counts were transform to area (hectares) based on the spatial resolution of the imagery. Classes lacking in a particular year were given a value of zero. The results were accumulated into a structured table documenting the year, class code, class name, and corresponding area.

Table 1 : Area statistics of LULC classes for 2016, 2020, and 2024.

Figure 6. Stacked bar chart displaying the proportional distribution of built-up land, cropland, tree cover, water, and bare land across the three study years.

Figure 7. Line graph showing temporal changes in area (km²) of major LULC classes between 2016, 2020, and 2024

4.4 Land use/Land cover (LULC) prediction using CA- Markov

The prediction of future land-cover patterns was carried out using the CA–Markov modelling approach. This method combines the statistical strength of Markov Chain analysis with the spatial neighborhood rules of Cellular Automata to simulate realistic landscape transitions.

4.4.1 Data Collection

LULC maps for 2016, 2020, and 2024, generated from the classification section, assisted as the main inputs for change detection and prediction. Several supplementary spatial variables were also prepared to support change potential modelling, including slope, distance from roads, distance from water, population density, and a built-up index.

4.4.2 Pre-processing of data

To ensure model compatibility, all datasets underwent standardized preprocessing. All raster layers were projected to a uniform coordinate reference system.

For improved data quality, categorical LULC maps were resampled using the nearest neighbor approach, while continuous data points were resampled using bilinear interpolation. To limit the data analysis to the study area, the Lahore district border was applied to both the LULC datasets and the added variables.

Continuous variables such as slope and distance raster's were normalized to a common scale (0–255) to ensure equal influence during model training. All raster datasets were aligned to the same spatial extent, grid origin, and resolution to enable pixel-level correspondence during the simulation process.

The MOLUSCE plugin in QGIS was used to create a shift matrix that assessed changes across LULC classes from 2016 to 2020. This matrix defined stable groups as well as classes that experienced gains or losses, which served as a framework for computing transition rates.

4.4.3 Transition Potential Modelling

MOLUSCE's neural network-based model was used for creating change potential maps. The inputs included the 2016 and 2020 LULC layers, as well as the additional geographical variables. The result was a set of probability surfaces that reflected the chance of class transitions over the landscape.

4.4.4 CA–Markov Simulation

The Markov Chain component assessed transition rates using observed changes between 2016 and 2020. Molecular automata were then used to incorporate spatial neighborhood

effects, resulting in a more realistic simulation of land-cover change. The number of simulation iterations fit the temporal gap between the baseline year and the predicted year.

4.4.5 Post-Processing of Outputs

Pixel counts from predicted maps were obtained and turned to hectares or square kilometers to calculate class-specific area statistics.

Table 3 : Class-wise Pixel Count and Area Estimates for the LULC Map of Lahore.

Class Value	Description	Pixel Count	Area (m ²)	Area (km ²)
1	Built-up	503158	4.53E+08	452.8422
2	Cropland	450847	4.06E+08	405.7623
3	Tree-cover	347901	3.13E+08	313.1109
4	Water	157976	1.42E+08	142.1784
5	Bare land	463960	4.18E+08	417.564

Predicted LULC maps were expressed symbolically to allow easy comparison across years. Change maps, graphs, and temporal plots were created to demonstrate shifts in land-cover classifications.

The envisioned LULC patterns were examined to identify important trends, such as urban growth, decreased vegetation cover, changes in water bodies, and changes in agricultural land. The results support long-term planning and oversight of the environment in the Lahore district.

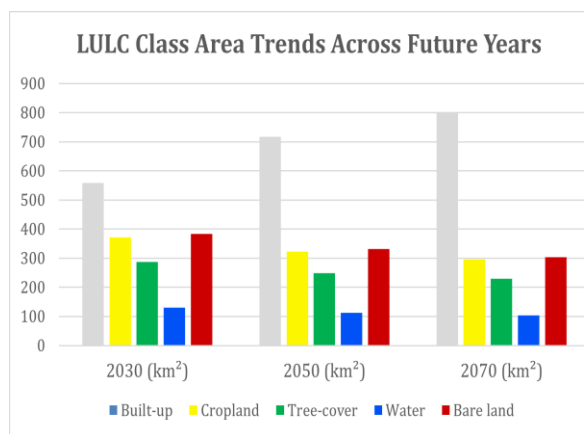


Figure 8. Pixel-class area distribution for the predicted 2030,2050,2070 LULC map, representing the expected spatial extent of each land-cover category based on the CA–Markov simulation using bar graph.

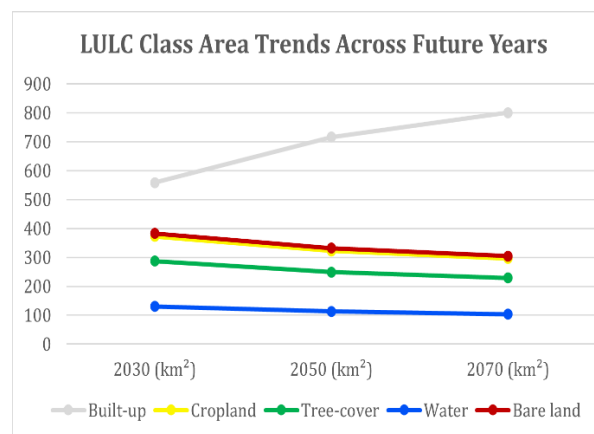
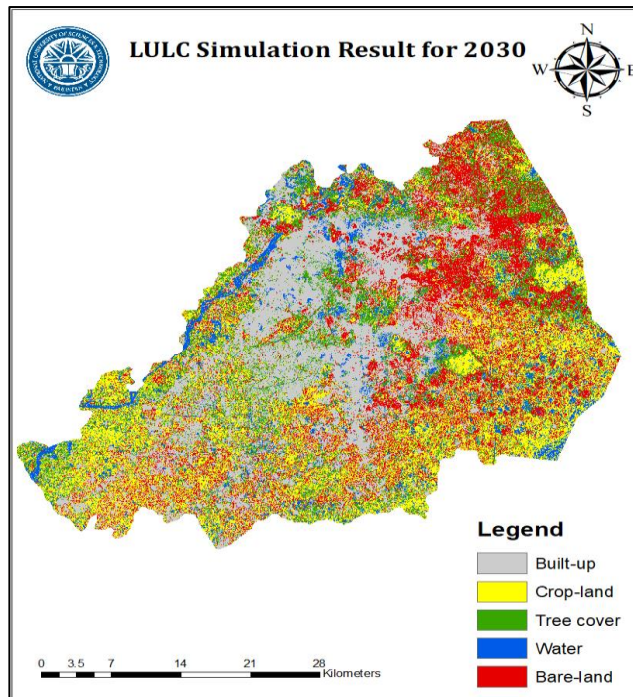
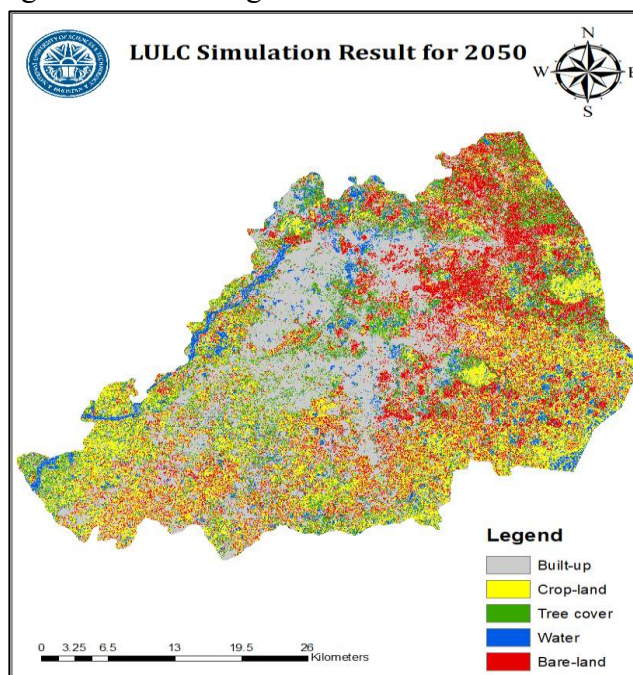


Figure 9. Pixel-class area distribution for the predicted 2030,2050,2070 LULC map, indicating the expected spatial extent of each land-cover category based on the CA–Markov simulation using line chart.



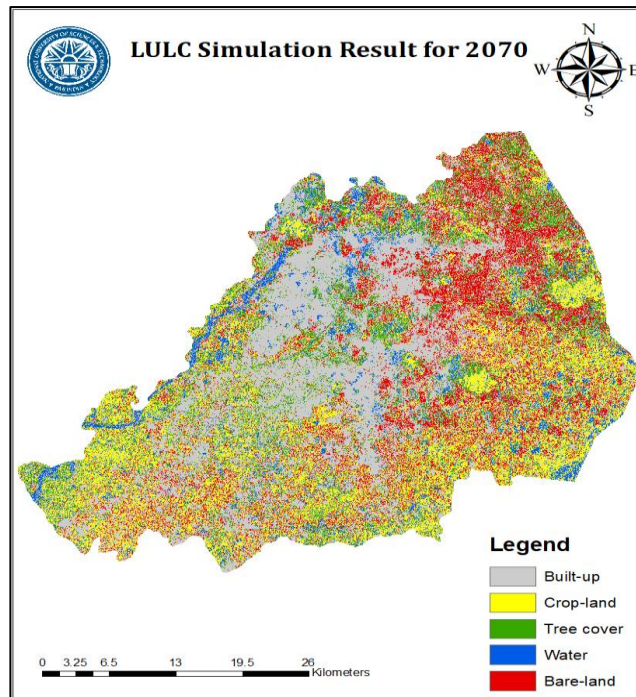
The 2030 prediction indicates moderate outer growth of built-up areas, especially along southern and western Lahore. Agricultural land and tree cover show original drops, while water and bare land remain fairly stable.

Figure 10. CA–Markov-based forecasted Land Use/Land Cover map for 2030 displaying early-stage urban growth and gradual drop in agricultural and vegetated land.



By 2050, built-up areas continue to expand hostilely, trading large portions of cropland and tree-covered zones. The urban trail becomes more continuous, especially around peripheral corridors. Vegetation and agricultural land show major contraction, reflecting strong urbanization trends.

Figure 11. CA–Markov simulated LULC map for Lahore for the year 2050, highlighting deepened urban growth and significant burden on cropland and natural vegetation.



The 2070 prognosis reveals a highly urbanized scene where built-up land becomes the central land cover class. Cropland and vegetated areas undertake severe lessening, and only imperfect blotches of natural cover remain. This scenario represents the long-term perpetuation of current development trends without ecological regulation.

Figure 12. Predicted Land Use/Land Cover distribution for 2070 based on long-term shift prospects, showing extensive urban supremacy and slight enduring natural land.

Table 4. Area statistics of predicted Land Use/Land Cover (LULC) classes for Lahore for the years 2030, 2050, and 2070 generated using the CA–Markov model

Class	2030 (km ²)	2050 (km ²)	2070 (km ²)
Built-up	558.5022	716.2245	800.2017
Cropland	372.429	322.2612	295.8912
Tree-cover	287.3655	249.0849	228.8556
Water	130.5702	113.0202	103.6593
Bare land	382.9806	331.8084	304.1487

5. Results

5.1 LULC Classification Results (2016, 2020, 2024)

The classification outputs for 2016, 2020, and 2024 expose clear spatial and temporal changes in Lahore’s landscape. Built-up areas show a incessant increase across all three years, designating rapid urban expansion driven by population growth and infrastructure development. In disparity, cropland shows prominent instabilities expanding between 2016 and 2020 but abating sharply by 2024. Tree cover declines between 2016 and 2020 but

recovers slightly in 2024, imitating localized plantation efforts and seasonal vegetation patterns. Water bodies show a extensive reduction, particularly from 2016 to 2020, followed by a minor growth in 2024. Bare land fluctuates throughout the study period but generally shrinkages, likely due to urban infill and conversion to built-up land.

The classified maps visually confirm these trends, with built-up spots mounting outward from the city center and agricultural zones lessening at the suburbs. These spatial patterns align with known urban growth pressures in Lahore.

5.2 Quantitative Change Analysis

Area statistics mined from the classified maps highlight the scale of change within each LULC class. Built-up land increased from 549.62 km² in 2016 to 600.32 km² in 2024, representing the fastest-growing category. Cropland unveiled the most variability, rising from 377.05 km² in 2016 to 475.69 km² in 2020 before falling to 370.93 km² in 2024. Tree cover decreased by approximately 42 km² between 2016 and 2020 but increased again to 433.90 km² in 2024. Water bodies declined significantly from 230.31 km² to 166.67 km² by 2020, consistent with reduced river flow and waterbody contraction. Bare land values also lifted modestly but show an overall downward trend.

These results indicate a consistent redeployment of land cover driven primarily by urban expansion and agricultural re-structuring.

5.3 CA–Markov Prediction Results (2030, 2050, 2070)

The CA–Markov model foresees continued growth of built-up land by 2030, convoyed by further reductions in cropland and tree-covered areas. Evolution potential maps show that urban expansion is most likely to occur along the city’s urban boundaries, predominantly in southern and western Lahore. Predicted maps reflect enduring pressure on agricultural fields and open land, suggesting that built-up land will remain the dominant class in the coming decade. Water bodies and vegetation show inadequate potential for recovery, signifying that without interposition, natural land covers may continue to decline. These prognoses emphasize the need for strategic planning to manage growth and protect remaining environmental resources.

The LULC prediction for 2050 exemplifies a noticeable prolongation of Lahore’s urban expansion. Compared to 2030, the built-up area breeds considerably, outspreading deeper into former agricultural regions. Cropland and tree-covered areas experience major reductions as urban development becomes more squashed and extents outward. The spatial pattern suggests that the southern and western edges of Lahore will spectator the highest conversion rates. Water bodies remain relatively stable, while bare land decreases due to urban infill processes.

The 2070 estimate validates an advanced stage of urban ascendancy across the Lahore district. The built-up class becomes the largest land category, inhabiting a generous portion of the landscape. Cropland and tree cover show severe drops, with natural vegetation persisting only in isolated outlying pockets. Urban expansion appears continuous and merged, reflecting highly exaggerated land conversion.

6. Accuracy Assessment

The accuracy assessment indicates excellent classification presentation for all study years. For 2016, the supervised LULC classification achieved a Kappa coefficient of 0.9910 and an overall accuracy of 99.28%. The 2020 classification performed even better with perfect scores (Kappa = 1.00, OA = 100%), representing a highly steadfast dissimilarity among land-cover classes. The Sentinel-2 classification for 2024 also produced very strong results (Kappa = 0.9909, OA = 99.28%). These values confirm the healthiness of the training samples and the suitability of both Landsat 8 and Sentinel-2 imagery for accurately mapping LULC patterns in Lahore

Table 5 : Accuracy assessment results for the supervised LULC classification for the years 2016, 2020, and 2024. Kappa coefficients and overall accuracies show steadily high classification reliability across all three years, with Landsat 8 (2016, 2020) and Sentinel-2 (2024) producing Kappa values above 0.99.

Year	Sensor	Overall Accuracy (%)	Kappa
2016	Landsat 8	99.28	0.9910
2020	Landsat 8	100	1.0000
2024	Sentinel-2	99.28	0.9910

7. Discussion

The results of this study highlight the rapid renovation of Lahore's landscape over the past decade. The sturdy increase in built-up area confirms that sprawl remains the prevailing driver of land-cover change. Similar patterns have been reported in previous studies on major Pakistani cities, where mounting housing schemes, commercial zones, and industrial areas continue to replace agricultural land.

The decline in cropland between 2020 and 2024 bring into line with the observed conversion of urban agricultural zones into housing developments. This trend raises critical concerns about long-term food security, groundwater recharge, and heat island escalation. The temporary reduction in tree cover followed by a bounce in 2024 may relate to recurrent vegetation cycles or targeted farmstead enterprises within the city.

Future predictions generated through the CA–Markov model emphasize these concerns. The anticipated extension of urban areas by 2030 indicates that, without preemptive urban planning, the remaining agricultural and natural areas will linger to shrink. This scenario highlights the urgency for policy measures pointed at balancing development needs with ecological sustainability.

By 2050, the decline of cropland and tree cover becomes more noticeable, indicating the evolution of peri-urban zones into unremitting built-up passages. This makeover aligns with the city's rising population density and strengthening of housing, transportation, and commercial infrastructure.

The 2070 results represent a critical verge where natural vegetation and agricultural land become severely lessened. This pattern validates how unfettered development could lead to environmental filth, reduced crop production, higher urban heat island effects, and increased pressure on water resources

Overall, the study demonstrates that Lahore's urban growth is occurring at the expenditure of environmental resources, demanding integrated land-management stratagems.

8. Conclusion

This study explored Land Use/Land Cover (LULC) changes in Lahore from 2016 to 2024 using Landsat 8 and Sentinel-2 imagery and predicted future landscape evolutions using the CA–Markov model. The multi-temporal classification results disclose a clear and continuous development of built-up land across the district, largely supplanting cropland, tree-covered areas, and open land. Water bodies displayed ample decline between 2016 and 2020, with only a minor recovery in 2024. These patterns reflect augmented urbanization driven by population growth, infrastructure development, and the pressure on peri-urban agricultural zones.

The accuracy assessment approves that the classification results are highly reliable. All three classified years confirmed exceptional performance, with overall accuracies exceeding 99% and Kappa coefficients above 0.99. These high accuracy scores validate the consistency of the training data, the appropriateness of the Random Forest classifier, and the reliability of the input satellite imagery. The strong accuracy further toughens confidence in both the temporal LULC analysis and the projections generated by the CA–Markov model.

Forthcoming prediction results for 2030 indicate a continual rise in urban expansion principally toward the southern and western parts of Lahore resulting in further deterioration of cropland and natural vegetation if no interference occurs. Without planned regulation and targeted safeguarding measures, the enduring natural areas may continue to shrink, leading to challenges related to environmental sustainability, groundwater recharge, agricultural productivity, and urban heat-island escalation.

The CA–Markov-based predictions for 2050 and 2070 indicate a future where Lahore becomes progressively urbanized, with built-up areas replacing most cropland, tree cover, and bare land. The 2050 projection highlights enhanced land conversion, while the 2070 scenario shows near-complete authority of urban land cover. These findings emphasize that, without deliberate environmental governance, Lahore may face severe ecological and socio-economic values, including declining agricultural yield, strengthened heat-island effects, loss of green spaces, and discriminating pressure on water and infrastructure systems.

Overall, this study highlights the vital need for tactical land-use planning and sustainable development plans to counter the rapid loss of environmental resources. The amalgamation of remote sensing and CA–Markov modelling offers a powerful approach for monitoring current land renovations and guiding long-term urban management decisions for Lahore.

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